

Tikhon Petrishchev

Embedded Systems Programmer, 22 years old, Moscow

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Key Skills

- Firmware design and development for real-time embedded systems (C, FreeRTOS, ESP32); reading and designing electrical schematics; development of electronic circuits and PCBs
- Web application development in Go and Python; Linux userspace administration and debugging; Docker
- Mobile application development, including multithreaded apps; Swift, SwiftUI
- Working with low-level protocols and binary data; reverse engineering for device integration; Bluetooth LE

Work Experience

IPat LLC, Moscow

June 2021 – Present

Project Manager, Architect, Lead Developer

August 2024 – October 2025

- Development of a specialized “Smart Pill Dispenser” integrated with a remote health monitoring and treatment adherence platform, funded by a grant from the Novosibirsk Academgorodok Science and Technology Park Foundation
- Circuit and prototype PCB development / KiCAD
  https://github.com/tikhonp/kicad-pill-dispenser
- Server-side development / Go, Echo, Templ, HTMX, SQLX, PostgreSQL
  https://github.com/tikhonp/medsenger-pill-dispenser-bot
- Firmware development / C, ESP-IDF, FreeRTOS, ESP32C3
  https://github.com/tikhonp/telepat-pill-dispenser
- 3D modeling of enclosures and internal components / SolidWorks

Lead Developer

2024

- Development of an API library for internal microservices in Go: https://github.com/tikhonp/maigo
- Integration of FreeStyle Libre glucose monitors / Go, Echo, Templ, SQLX, PostgreSQL
- Integration of Xiaomi smart scales with the Medsenger remote monitoring service. Flutter-based mobile app for iOS and Android. Integration microservice / Go, Echo, Templ, SQLX, PostgreSQL

Lead Developer

2023

- Development of a microservice for configurable automated medical phone surveys via the Voximplant service / Python, Django, Django Rest Framework, PostgreSQL

Project Manager, iOS Developer

2023

- Development of a mobile app integrating with the Axma Hemocard Bluetooth LE tonometer and ECG device / Swift, SwiftUI, Kotlin

Project Manager, iOS Developer

2022 – 2024

- Development of the Medsenger iOS mobile app (~60k lines) / Swift, SwiftUI, CoreData, HealthKit, WebRTC video calls

Reverse Engineer, iOS Developer

2022

- Integration of the “Contec” Bluetooth spirometer into the medical messenger; reverse engineering of the Bluetooth LE protocol
- Development of an iOS mobile application for the Contec spirometer / Swift, SwiftUI, CoreBluetooth
- Backend for the integration microservice using Flask

iOS Developer

2021

- Integration of the “Serdechko” cardiograph into the Medsenger — medical messenger
- Patching the mobile app on iOS and Android
- Backend for the microservice using Flask

Project Author, Developer

2019–2021

- Embedded software for a smart speaker integrated with a medical service; development of a library for Yandex SpeechKit / Python, Yandex SpeechKit / https://github.com/tikhonp/yandex-speechkit-lib-python
- Backend / Django, Django Rest Framework, PostgreSQL
- iOS application / Swift, SwiftUI

Education

RTU MIREA, Moscow

Bachelor’s degree, Class of 2026
Industrial Informatics

Achievements

Authorship of Computer Programs

- Certificate of state registration of a computer program No. 2024611961 “Medsenger.Prenatal – subsystem for remote monitoring and informational support for pregnant women” dated January 26, 2024
- Certificate of state registration of a computer program No. 2024611962 “Medsenger.Hyper – subsystem for remote blood pressure monitoring using various measurement data transfer methods” dated January 26, 2024
- Certificate of state registration of a computer program No. 2024616051 “Oncofertility Registry” dated March 15, 2024

Competitions

- Winner of the YoHack hackathon by Croc, April 2020

Interests and Hobbies

- Classical and jazz music, piano
- Sport orienteering, running